

Leonardo Bonanno

Department of Mathematics, University of Chicago
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Citizenship: United States

Education

Ph.D. in Mathematics, University of Chicago 2023–present
Advisor: Ewain Gwynne.
Expected completion: Spring 2028.

M.S. in Mathematics, University of Chicago 2025

B.S. in Mathematics with Honors, Stanford University 2019–2023
Minor in Statistics.
Honors thesis: *Pseudodifferential Operators and the Hodge Decomposition Theorem*.

Research interests

Probability Theory, Random Geometry in Statistical Mechanics

Topic examination manuscript. *Parameterized walk on orthogonal tilings converges to Brownian motion*. Available upon request.

Teaching

University of Chicago, Department of Mathematics

Instructor

MATH 15100: Calculus I	Autumn 2026
MATH 13200: Elementary Functions and Calculus II	Winter 2026
MATH 13100: Elementary Functions and Calculus I	Autumn 2025

Graduate Teaching Assistant

MATH 23500: Markov Chains, Martingales, and Brownian Motion	Spring 2026
MATH 23500: Markov Chains, Martingales, and Brownian Motion	Spring 2025
MATH 27000: Basic Complex Variables	Winter 2025
MATH 23500: Markov Chains, Martingales, and Brownian Motion	Autumn 2024

Stanford University, Department of Mathematics

2020–2023

Grader

Multivariable Calculus (Winter 2023); Introduction to Analysis (Spring 2022); Functional Analysis (Winter 2022); Graduate Analysis (Autumn 2021); Group Theory (Summer 2021); Measure Theory (Spring 2021).

Talks and posters

Parameterized Random Walks on Orthogonal Tilings July 7, 2026
IMSI, Chicago (poster).

Mentoring and outreach

University of Chicago Mathematics REU Summer 2024–2025
Research Mentor

Mentored undergraduate students in the Research Experience for Undergraduates program.

University of Chicago Directed Reading Program Autumn 2024
Mentor

Mentored an undergraduate student through an independent mathematics reading project.

Euler Circle Winter 2022–Spring 2023
Volunteer Teaching Assistant

Gems of Linear Algebra and Combinatorics (Winter 2023); Number Theory and Group Theory (Autumn 2022); Differential Geometry (Spring 2022); Real Analysis (Winter 2022).

Employment

Amazon Web Services June–September 2021
Software Engineering Intern, AWS X-Ray Voyager Team

Stanford University, Department of Statistics June–September 2020
Undergraduate Researcher, Palacios Lab